

Code Based Math Problems with Solutions

1. Chunxiang is 4 years old now. His younger brother is half his age. When Chunxiang is 50 years old, how old will his younger brother be?

Solution. Chunxiang is 4, so his younger brother is 2. The age difference is 2 years. Therefore, when Chunxiang is 50, his younger brother will be $50 - 2 = 48$ years old. Hence, the answer is 48.

2. A family has two children. Given that at least one of them is a boy, what is the probability that the other child is also a boy?

Solution. The possible cases are BB , BG , and GB . Out of these 3 equally likely cases, only 1 case has both children as boys. Therefore, the probability is $\frac{1}{3}$.

3. Using exactly 6 matchsticks, what is the maximum number of equilateral triangles you can form?

Solution. If we allow a three-dimensional construction, the 6 matchsticks can form a regular tetrahedron, which has 4 equilateral triangular faces. Hence, the maximum number is 4.

4. For the four lattice polygons shown in the figure, find

$$Area_1 + Area_2 + Area_3 + Area_4.$$

Solution. Using Pick's Theorem (or direct counting), the four areas are

$$Area_1 = 9, \quad Area_2 = 6, \quad Area_3 = 8.5, \quad Area_4 = 11.$$

Therefore,

$$Area_1 + Area_2 + Area_3 + Area_4 = 9 + 6 + 8.5 + 11 = 34.5.$$

Thus, the answer is 34.5.

5. If the tree, when it falls, forms a right triangle with the ground, will the boy be safe? If not, calculate how many feet the boy needs to move backward.

Solution. The falling part of the tree has length 8 ft, and the break point is 4 ft above the ground. Thus the horizontal reach is

$$\sqrt{8^2 - 4^2} = \sqrt{64 - 16} = \sqrt{48} = 4\sqrt{3} \approx 6.93 \text{ ft.}$$

Since the boy is only 6 ft away, he is **not** safe. He must move backward by

$$4\sqrt{3} - 6 \approx 0.93 \text{ ft.}$$

Hence, the boy is not safe, and he needs to move back 0.93 ft (approximately).

6. Find:

$$\sqrt{1.77777\ldots}, \quad \sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \cdots}}}}, \quad \sqrt{2\sqrt{4\sqrt{8\sqrt{16\sqrt{\cdots}}}}}.$$

Solution. For the first one, let

$$x = \sqrt{1.77777\ldots} = \sqrt{\frac{16}{9}},$$

so

$$x = \frac{4}{3}.$$

For the second one, let

$$x = \sqrt{1 + x}.$$

Then

$$x^2 = 1 + x \quad \Rightarrow \quad x^2 - x - 1 = 0,$$

so

$$x = \frac{1 + \sqrt{5}}{2}.$$

For the third one, let

$$x = \sqrt{2\sqrt{4\sqrt{8\sqrt{16\sqrt{\cdots}}}}}.$$

Then

$$x = \sqrt{2x_1}, \quad x_1 = \sqrt{4x_2}, \quad x_2 = \sqrt{8x_3}, \dots$$

The pattern gives the fixed value $x = 4$.

Therefore, the answers are

$$\boxed{\frac{4}{3}}, \quad \boxed{\frac{1 + \sqrt{5}}{2}}, \quad \boxed{4}.$$

7. Complete the missing shapes in the figure and identify the reversed mathematical terms.

Solution. Using mirror symmetry about the central vertical line, the missing shapes (from top to bottom) are:

up-left arrow, left flag, backslash , right flag, right arrow.

The reversed mathematical terms are:

EIGENSPACE

ISOMORPHISM

SURJECTION

INVOLUTION

MANIFOLD

IDEMPOTENT

STOCHASTIC

HOMEOMORPHISM

DIFFEOMORPHISM

8. Evaluate:

$$\sin^2 0^\circ + \sin^2 1^\circ + \cdots + \sin^2 89^\circ + \sin^2 90^\circ$$

and

$$100 - 98 + 96 - 94 + 92 - 90 + \cdots + 8 - 6 + 4 - 2.$$

Solution. Using the identity

$$\sin^2 \theta + \sin^2(90^\circ - \theta) = 1,$$

the terms pair up to give

$$45 + \sin^2 90^\circ = 45 + 1/2 = \frac{91}{2}.$$

So,

$$\sin^2 0^\circ + \sin^2 1^\circ + \cdots + \sin^2 90^\circ = \boxed{\frac{91}{2}}.$$

For the second sum, group the terms:

$$(100 - 98) + (96 - 94) + \cdots + (4 - 2).$$

Each pair equals 2, and there are 25 pairs. Hence the total is

$$25 \times 2 = 50.$$

So the answer is $\boxed{50}$.

9. Find the fallacy in each of the three arguments shown.

Solution.

(a) In the first argument, the fallacy occurs when going from

$$(3 - x)^2 = (x - 3)^2$$

to

$$3 - x = x - 3.$$

Since equal squares do *not* imply the numbers themselves are equal in sign, this step is invalid.

(b) In the second argument, the fallacy occurs when dividing by

$$a^2 - ab = a(a - b).$$

But since $a = b$, we have $a - b = 0$, so this is division by zero.

(c) In the third argument, the fallacy occurs when simplifying square roots incorrectly. In general,

$$\sqrt{x^2} = |x|,$$

not simply x for all real x .

10. There are 9 identical-looking coins, and exactly 1 of them is fake (it has a different weight, but you do not know whether it is heavier or lighter than a real coin). Using a balance scale, what is the minimum number of weighings needed to guarantee finding the fake coin and determining whether it is heavier or lighter?

Solution. Each weighing has 3 possible outcomes: left side heavier, right side heavier, or balanced. Thus, with n weighings, we can distinguish at most 3^n cases.

Here there are 9 choices for the fake coin, and for each coin it may be either heavier or lighter, so there are

$$9 \times 2 = 18$$

possible cases.

Since

$$3^2 = 9 < 18, \quad 3^3 = 27 \geq 18,$$

at least 3 weighings are necessary, and 3 weighings are also sufficient. Therefore, the minimum number is $\boxed{3}$.

11. A palindrome poem reads the same forward and backward. Suppose a 7-letter word follows a symmetrical pattern like

$$A B C D C B A$$

(with the 1st and 7th letters the same, etc.). If you only use the 26 standard English letters, how many different 7-letter combinations can be formed using this pattern?

Solution. The pattern is determined by the first 4 letters:

$$A, B, C, D.$$

Each can be chosen in 26 ways, so the total number is

$$26^4 = 456,976.$$

Hence, the answer is $\boxed{456,976}$.

12. A housing estate has 300 apartments, and each door uses a 4-digit passcode (from 0000 to 9999, so there are 10,000 possible passcodes). What is the probability that at least two apartments share the exact same passcode?

Solution. This is a birthday-problem type question. The probability that all 300 pass-codes are different is

$$\frac{10000}{10000} \cdot \frac{9999}{10000} \cdot \frac{9998}{10000} \cdots \frac{9701}{10000} = \prod_{k=0}^{299} \left(1 - \frac{k}{10000}\right).$$

Therefore,

$$P(\text{at least one match}) = 1 - \prod_{k=0}^{299} \left(1 - \frac{k}{10000}\right) \approx 0.989224.$$

So the probability is approximately 98.92%.

- 13.** Nine local working-class individuals are sitting in a café in Malaysia. Each of them earns an average annual salary of RM 60,000. Suddenly, a top C-Suite executive earning RM 12,000,000 per year walks in and sits down with them. What happens to the mean annual salary of the 10 people in the café? Does the median annual salary of the 10 people change significantly?

Solution. The total income of the 9 workers is

$$9 \times 60,000 = 540,000.$$

Adding the executive gives

$$540,000 + 12,000,000 = 12,540,000.$$

So the new mean is

$$\frac{12,540,000}{10} = 1,254,000.$$

Thus, the new mean annual salary is RM 1,254,000.

The median, however, stays around the middle of the workers' salaries, so it remains approximately RM 60,000. Hence, the mean changes drastically, but the median does not change significantly.

- 14.** In a city of 1,000 people, only 1 person has a rare disease. A hospital test detects the sick person correctly, but gives a false positive to 1% of healthy people. If a patient tests positive, what is the simplest mathematical step a doctor can take to be almost 100% sure the patient is truly sick?

Solution. Among 1,000 people:

$$1 \text{ is sick, } \quad 999 \text{ are healthy.}$$

The sick person tests positive, and about

$$0.01 \times 999 \approx 10$$

healthy people also test positive falsely. So there are about 11 positive tests in total, but only 1 is truly sick. Therefore,

$$P(\text{sick} \mid \text{positive}) \approx \frac{1}{11} \approx 9.1\%.$$

The simplest step is to **repeat the test independently**. If the patient tests positive multiple times in a row, the chance of repeated false positives becomes extremely small. Thus, the practical answer is:

Do repeated independent confirmatory tests.

- 15.** A delivery robot needs to travel from Point A (Start) to Point B (Destination) through the road network shown. Which route gives the shortest total time, and what is the minimum time needed?

Solution. From the figure, the available routes are:

$$A \rightarrow C \rightarrow B : \quad 2 + 1 = 3$$

$$A \rightarrow D \rightarrow C \rightarrow B : \quad 5 + 1 + 1 = 7$$

$$A \rightarrow D \rightarrow B : \quad 5 + 3 = 8$$

So the shortest route is

$$A \rightarrow C \rightarrow B$$

with minimum total time

$$3 \text{ minutes}.$$

- 16.** In the illustrated Mondrian Art Problem, determine the minimum score, where

$$\text{Score} = \text{Area of Largest Rectangle} - \text{Area of Smallest Rectangle}.$$

Solution. For the illustrated example, the largest rectangle has area 10 and the smallest rectangle has area 5. Therefore,

$$\text{Score} = 10 - 5 = 5.$$

Hence, the minimum score shown is 5.

- 17.** Fill each row with an equation that is equal to 45.

Solution. One possible set of six valid equations is:

$$50 - 5 = 45$$

$$40 + 5 = 45$$

$$90 \div 2 = 45$$

$$15 \times 3 = 45$$

$$54 - 9 = 45$$

$$36 + 9 = 45$$

So these are acceptable answers.

18. Use the given numbers and operations to make 24.

Solution. A correct expression is

$$\frac{12 \times 10}{6 - 1} = 24.$$

Hence, the answer is

$$\boxed{\frac{12 \times 10}{6 - 1} = 24}.$$

19. Solve the cryptogram puzzle.

Solution. There is no single fixed answer here, because the PDF only provides a link to an online cryptogram puzzle. The answer depends on the specific cryptogram generated at that time. Hence, the correct response is:

No fixed answer; it depends on the actual puzzle shown online.

20. In the Monty Hall Problem, you pick 1 of 3 doors. Behind one door is a car, and behind the other two are goats. After you choose, the host opens one of the other two doors to reveal a goat, and then offers you the chance to switch. Does switching increase your chance of winning the car? What is the probability of winning if you switch?

Solution. Yes, switching increases your chance of winning. Initially, your chosen door has probability

$$\frac{1}{3}$$

of hiding the car, while the other two doors together have probability

$$\frac{2}{3}.$$

When the host reveals a goat, that full $\frac{2}{3}$ probability transfers to the remaining unopened door. Therefore,

$$P(\text{win by switching}) = \boxed{\frac{2}{3}}, \quad P(\text{win by staying}) = \boxed{\frac{1}{3}}.$$

So you should switch.